Edge Cases

Prompt

Act as a Software Architect and identify edge cases for the Inventory Management System (IMS), covering:

- Stock additions, removals, concurrent updates, and stock consistency
- Handling obsolete, invalid, and duplicate SKUs
- Data validation, boundary conditions, and failure scenarios
- Other relevant business and technical edge cases

Response

Edge Cases for Inventory Management System

Below are important **edge cases**, failure scenarios, and validation conditions that should be considered while designing and testing the Inventory Management System.

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1. Stock Update Edge Cases

1.1 Negative Inventory

Scenario

User tries to remove more stock than available.

Example

Available stock = 5
Requested removal = 10

Expected Behavior

- System should reject transaction
- Display insufficient stock message
- Log failed attempt

1.2 Concurrent Stock Updates

Scenario

Two users update the same product inventory simultaneously.

Example

- User A dispatches 5 units
- User B dispatches 7 units
- Available stock = 10

Risk

Race condition causing incorrect stock.

Expected Behavior

- Use transaction locking/versioning
- Prevent inconsistent inventory
- Only one transaction succeeds if stock becomes insufficient

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1.3 Duplicate Stock Submission

Scenario

User clicks “Submit” multiple times due to slow network.

Expected Behavior

- Prevent duplicate transactions
- Use idempotency tokens or request locking
- Only one stock movement should be recorded

1.4 Stock Update During System Failure

Scenario

Database/network failure occurs mid-transaction.

Expected Behavior

- Entire transaction should rollback
- No partial stock updates
- Error should be logged

1.5 Inventory Count Becomes Decimal

Scenario

Inventory quantity becomes fractional unexpectedly.

Example

Stock = 5.5 for a non-divisible product

Expected Behavior

- Validate unit type
- Allow decimals only for supported units (kg, liters, etc.)

1.6 Stock Adjustment Without Reason

Scenario

User manually changes inventory.

Expected Behavior

- Adjustment reason mandatory
- Audit log required
- Optional approval workflow

1.7 Backdated Inventory Transactions

Scenario

User enters stock transaction with old date.

Risks

- Incorrect historical reporting
- Ledger inconsistencies

Expected Behavior

- Restrict backdated entries
- Require admin approval
- Recalculate inventory ledger carefully

1.8 Timezone Issues

Scenario

Warehouses in different regions update stock.

Expected Behavior

- Store timestamps in UTC
- Display localized time per user

1.9 Offline Stock Updates

Scenario

Warehouse device loses internet connectivity.

Expected Behavior

- Queue offline transactions
- Sync when online
- Resolve conflicts safely

1.10 Bulk Upload Failure

Scenario

CSV upload partially succeeds.

Expected Behavior

- Validate entire file before processing

- Provide row-level errors
- Support rollback

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2. SKU Edge Cases

2.1 Duplicate SKU Creation

Scenario

Two products created with same SKU.

Expected Behavior

- Enforce unique SKU constraint
- Reject duplicate SKU creation

2.2 SKU Case Sensitivity

Scenario

abc123 and ABC123 treated differently.

Expected Behavior

- Normalize SKU casing
- Use case-insensitive uniqueness

2.3 Invalid SKU Format

Scenario

SKU contains unsupported characters.

Example

SKU = INV#@!

Expected Behavior

- Validate SKU format
- Restrict special characters if required

2.4 SKU Reuse After Product Deletion

Scenario

Deleted product SKU reused accidentally.

Risks

- Historical reporting conflicts
- Audit inconsistencies

Expected Behavior

- Soft delete products
- Prevent SKU reuse or archive old SKU permanently

2.5 Very Long SKU Values

Scenario

SKU exceeds database/UI limits.

Expected Behavior

- Validate max length
- Prevent truncation issues

2.6 Barcode Collision

Scenario

Two products mapped to same barcode.

Expected Behavior

- Barcode uniqueness validation
- Reject duplicate barcode assignment

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3. Removing Older / Invalid / Duplicate SKU Edge Cases

3.1 Removing Product With Existing Transactions

Scenario

User deletes product already used in inventory history.

Expected Behavior

- Prevent hard delete
- Mark product inactive/archive
- Preserve transaction history

3.2 Deleting Active Product With Existing Stock

Scenario

Product still has inventory.

Expected Behavior

- Prevent deletion until stock = 0
- Require stock clearance workflow

3.3 Removing Product Linked to Purchase Orders

Scenario

Product exists in active PO.

Expected Behavior

- Restrict deletion
- Require PO closure/cancellation first

3.4 Removing Duplicate SKUs During Data Migration

Scenario

Legacy system contains duplicates.

Expected Behavior

- Detect duplicates during import
- Generate exception report
- Require manual resolution

3.5 Invalid Historical SKUs

Scenario

Old products contain deprecated SKU formats.

Expected Behavior

- Maintain backward compatibility
- Allow historical lookup
- Prevent new invalid formats

3.6 Product Merge Scenario

Scenario

Two duplicate products need merging.

Expected Behavior

- Consolidate inventory safely
- Merge transaction history carefully
- Preserve audit logs

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4. Inventory Workflow Edge Cases

4.1 Partial Goods Receiving

Scenario

Supplier delivers partial quantity.

Expected Behavior

- Allow partial receipt
- Keep PO open
- Update pending quantity

4.2 Over Receiving

Scenario

Received quantity exceeds PO quantity.

Expected Behavior

- Restrict or require approval
- Log variance

4.3 Dispatch From Wrong Warehouse

Scenario

Stock exists globally but not in selected warehouse.

Expected Behavior

- Validate warehouse-level inventory

4.4 Simultaneous Transfer & Dispatch

Scenario

Transfer and dispatch happen at same time.

Expected Behavior

- Lock inventory records
- Prevent double deduction

4.5 Inventory Cycle Count Variance

Scenario

Physical stock differs from system stock.

Expected Behavior

- Support reconciliation workflow
- Record variance reason
- Require approval

4.6 Expired Inventory Dispatch

Scenario

Expired product dispatched accidentally.

Expected Behavior

- Block dispatch
- Generate expiry alerts

4.7 Unit Conversion Errors

Scenario

Stock managed in multiple units.

Example

Boxes vs individual units

Expected Behavior

- Maintain conversion rules
- Prevent rounding inconsistencies

4.8 Return Quantity Exceeds Original Dispatch

Scenario

Customer returns more items than purchased.

Expected Behavior

- Restrict excessive returns
- Validate against original transaction

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5. Security & Audit Edge Cases

5.1 Unauthorized Inventory Modification

Scenario

User bypasses UI and calls API directly.

Expected Behavior

- Backend permission validation mandatory

5.2 Audit Log Tampering

Scenario

Admin tries to edit logs.

Expected Behavior

- Audit logs immutable
- Separate secure logging storage

5.3 Session Timeout During Transaction

Scenario

User session expires while updating inventory.

Expected Behavior

- Prevent partial save
- Preserve draft if applicable

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6. Reporting Edge Cases

6.1 Historical Inventory Recalculation

Scenario

Backdated transaction changes past stock.

Expected Behavior

- Recalculate historical balances correctly

6.2 Large Data Volume

Scenario

Millions of inventory transactions.

Expected Behavior

- Pagination
- Indexed queries
- Async report generation

6.3 Report Export Failure

Scenario

Large Excel/PDF export crashes.

Expected Behavior

- Background export job
- Download notification

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7. Data Migration Edge Cases

7.1 Duplicate Legacy Records

Expected Behavior

- Duplicate detection rules
- Merge strategy

7.2 Missing Mandatory Fields

Expected Behavior

- Validation report
- Reject incomplete records

7.3 Invalid Historical Inventory Balances

Expected Behavior

- Migration reconciliation process
- Exception reporting

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8. High-Risk Scenarios to Test Thoroughly

Area	Critical Risk
Concurrent stock updates	Inventory corruption
Duplicate SKU handling	Product ambiguity
Backdated transactions	Reporting inconsistencies
Stock transfer race conditions	Double deduction
Transaction rollback failures	Incorrect inventory
Inventory reconciliation	Financial mismatch
Bulk uploads	Mass incorrect updates
Warehouse-level validation	Wrong stock visibility
Audit logging	Compliance failure
Soft delete handling	Historical data loss